

LECTURE 8: TESTING

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REFERENCE CB, Chapter 8.1, 8.3.1, and 8.3.4–5, or HMC, Chapter 4.5–4.6, H22, Chapter 13.

1. HYPOTHESES

A hypothesis is a statement about the population distribution that generated the data. Given a statistical model $\{F_X(\cdot \mid \theta) : \theta \in \Theta\}$ for the data X , we can partition the parameter space Θ into two sets, Θ_0 and Θ_1 , such that if $\theta \in \Theta_0$, the hypothesis is true; otherwise it is false. Typically, the hypothesis is called the *null hypothesis*, and we write $H_0 : \theta \in \Theta_0$. The class of alternatives (also called the *alternative hypothesis*) is written as $H_1 : \theta \in \Theta_1$. If a hypothesis uniquely identifies the distribution of the data—meaning the parameter space under that hypothesis contains only a single element—it is called *simple*. Otherwise, the hypothesis is called *composite*.

Example 1. Let $X_1, \dots, X_n$ be a random sample from distribution $\mathcal{N}(\theta, \sigma^2)$ with σ^2 known, and $\theta \in \Theta$. If $\Theta = \mathbb{R}$, and $\Theta_0 = \{\theta : \theta \leq \theta_0\}$, then $H_0 : \theta \leq \theta_0$ and $H_1 : \theta > \theta_0$. Both hypotheses are composite. $\boxtimes$

Example 2. Let $\Theta_0 = \{\theta_0\}$, so that $H_0 : \theta = \theta_0$ and let $H_1 : |\theta - \theta_0| > d$. Here the null is simple, and the alternative is composite, and separated from the null by distance d . For instance, in a randomized controlled trial, θ may correspond to the treatment effect, in which case the null with $\theta_0 = 0$ corresponds to the hypothesis of no treatment effect, and d corresponds to a “minimum detectable effect”, the smallest treatment effect that is of substantive interest. Here the set $\{\theta : 0 < |\theta - \theta_0| \leq d\}$ is not part of the parameter space. A one-sided version of this setup is to test $H_0 : \theta \leq \theta_0$ against $H_1 : \theta \geq \theta_0 + d$, and both the null and the alternative are composite. $\boxtimes$

It is good practice to be explicit about both the null and the alternative hypotheses, since this makes any restrictions on the full parameter space Θ clear.

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2. TESTING

Testing a hypothesis means that we need to decide, based on the value of a random variable X , whether some hypothesis H_0 that has been formulated is correct. We choose between two decisions: *accept* or *reject* the hypothesis, so that, viewed as a decision problem, the action space is $\mathcal{A} = \{\text{accept}, \text{reject}\}$. (Some people argue that “do not reject” is perhaps more accurate).

A non-randomized decision rule splits the sample space $\mathcal{X}$ into two parts, S_0 and S_1 , the acceptance and *critical* regions. If we observe $X \in S_1$, then we reject H_0 , otherwise we accept it. If we denote the decision to reject H_0 as “1”, then we can view the decision rule as a function $\phi: \mathcal{X} \mapsto \{0, 1\}$. If S_1 is the critical region, then the function is given by $\phi(x) = \mathbb{1}\{x \in S_1\}$. Typically, the critical region can be written as $S_1 = \{T(X) \geq c\}$ for some *test statistic* T and a cutoff $c \in \mathbb{R}$, also called a *critical value*. In such cases,

$$\phi(X) = \mathbb{1}\{T(X) \geq c\}. \quad (1)$$

Since we can take two actions, and the hypothesis is either true or false, a testing problem has four possible outcomes:

	H_0 true	H_1 true
Accept	Correct decision	Type II error
Reject	Type I error	Correct decision

For any $\theta \in \Theta$, let

$$\pi_\phi(\theta) = E_\theta[\phi(X)]$$

denote the *power function* of the test ϕ . Often we drop the subscript, and just write $\pi(\theta)$. The probability of making a Type I error when $\theta_0 \in \Theta_0$ is the true value is $\pi(\theta_0)$. The probability of making a Type II error when $\theta_1 \in \Theta_1$ is the true value is $1 - \pi(\theta_1)$.

Ideally, we’d like to keep both types of error small. We can always reduce the probability of a type I error by decreasing $\phi(X)$, though this inherently increases the probability of a type II error. For instance, for tests of the form in eq. (1), making c big reduces type I error rate, but increases type II error rate. So there is a trade-off.

The consequences of type I and type II errors are often quite different, so it makes sense to treat them asymmetrically. A principled way of doing this is to set up the problem as a decision problem, and assign loss 1 to a type I error and loss L to a type II error. Then the risk is

$$R(\theta, \phi) = \mathbb{1}\{\theta \in \Theta_0\} \pi_\phi(\theta) + L \mathbb{1}\{\theta \in \Theta_1\} (1 - \pi_\phi(\theta)).$$

The maximum risk is then given by $\bar{R}(\phi) = \max\{\sup_{\theta \in \Theta_0} \pi_\phi(\theta), L \sup_{\theta \in \Theta_1} (1 - \pi_\phi(\theta))\}$. The first argument, the maximum null rejection probability, is called the *size* of the test.

Similar to estimation, a principled approach for selecting among admissible tests is

to use a minimax test, ϕ_L^* , one that minimizes the maximum risk if we place loss L on the type II error. For the test to be minimax, the size must equal $L \sup_{\theta \in \Theta_1} (1 - \pi_\phi(\theta))$. Otherwise, if, say, the size was smaller than the maximum risk under the alternative, we could adjust the test to reject more often and thus reduce the maximum risk. Suppose that the minimax risk is α . Then it must be that the size of the test is α , and the minimum power of the test is $1 - \alpha/L$.

In practice, rather than choosing L , it is customary to instead impose a bound $\alpha \in (0, 1)$, called the *level of significance*, on the risk we're willing to accept under the null. That is, we require that

$$\sup_{\theta \in \Theta_0} E_\theta[\phi(X)] \leq \alpha. \quad (2)$$

In other words, we require that the *size of the test may not exceed a given level of significance*. Many people use the terms “size” and “level” loosely and sometimes interchangeably. Subject to the level constraint, we maximize the minimum power under the alternative. The resulting test is called “maximin”. Note the maximin test is equivalent to the minimax test, for a particular choice of L . Specifically, if the minimum power of the maximin test is $\underline{\pi}$, and we set $L = \alpha/(1 - \underline{\pi})$, then the maximin test is the same test as the minimax test ϕ_L^* .

Example 2 (continued). In randomized controlled trials, it is customary to set the level of significance to be 5%. The sample size and the minimum detectable effect are specified so that it is possible to construct a test with 80% power. This is equivalent to setting $L = 0.05/0.20 = 1/4$, meaning rejecting a null effect is four times as costly as failing to detect a true effect of magnitude d or bigger. See Armstrong, Kitagawa, and Tetenov (2026) for more discussion. $\square$

Remark 1. While it is by far more common to impose a level constraint and leave the choice of L implicit, it is not clear this is a wise default. For example, Lehmann (1958) argues that rather than maximizing power subject to a level constraint, without regard to what power may be achievable, “it seems more natural when deciding on a significance level (and this suggestion is certainly not new) to take into account also what power can be achieved with the given experiment.” Nonetheless, we go along with this default, and focus on maximin, rather than minimax tests. Since a minimax test is maximin for a particular choice of level constraint α , we don't lose any generality in doing this.

The minimax/maximin approach requires separating the null and the alternative. This is because the power function is, under mild regularity conditions, always continuous, so without the separation the minimal power and the test size coincide. While in general, the form of the maximin test may depend on the choice of the minimum detectable effect d , there are important settings in which this is not the case, and the maximin test is the same no matter what d is. In such cases, the test is in fact *uniformly most powerful*: it is not possible to construct another test at the same level of significance that has better power for some $\theta \in \Theta_1$. We discuss these settings and how to find uniformly most powerful tests in the next lecture.

Recall from our earlier decision theory discussion that it is common to restrict the class of decision rules *ex ante*, and that two most common ways of doing so are to impose unbiasedness, or to appeal to the invariance principle. We won't discuss invariance, but we will consider unbiasedness in the next lecture. These restrictions can be used in conjunction with the maximin approach. As we discussed in earlier lectures, an alternative to the minimax/maximin approach is to use average risk. In the testing context, this yields the Bayesian tests discussed in the previous lecture.

In this lecture, we focus on another method for finding tests, called the *ad hoc* method.

Example 1 (continued). Suppose we want a test with level 5%. By a sufficiency argument, we can restrict attention to tests that depend on the data only through the sufficient statistic $\bar{X}_n$. Furthermore, it seems reasonable to consider tests of the form $\phi(X) = \mathbb{1}\{\bar{X}_n \geq c\}$ for some c . We need to choose c such that

$$P_\theta(\bar{X}_n \geq c) = P_\theta\left(\frac{\bar{X}_n - \theta}{\sigma/\sqrt{n}} \geq \frac{c - \theta}{\sigma/\sqrt{n}}\right) = 1 - \Phi(\sqrt{n}(c - \theta)/\sigma) \leq 0.05,$$

whenever $\theta \leq \theta_0$. Since for a fixed c , the power function is increasing in θ (why?), we just need to make sure that the constraint $1 - \Phi(\sqrt{n}(c - \theta_0)/\sigma) \leq 0.05$ holds. This rejection probability is decreasing in c . Hence, we can pick c such that this constraint holds with equality. This gives $c = \sigma\Phi^{-1}(0.95)/\sqrt{n} + \theta_0$. So, our test will reject the null whenever the z-statistic,

$$Z = \frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}},$$

exceeds $\Phi^{-1}(0.95)$: $\phi(X) = \mathbb{1}\{Z \geq \Phi^{-1}(0.95)\}$. The test is known as a (one-sided) z-test. $\square$

2.1. *p-value*

In hypothesis testing, we either accept or reject the null, given a prespecified significance level. Sometimes it may be of interest to know the sensitivity of the decision to the choice of significance level. The concept of a *p-value* gives us such a sensitivity measure: it is the smallest significance level for which we would still reject the null.

Remark 2. This definition is a little different from the definition on p. 252 in HMC, or Section 8.3.2 in CB, but it is more precise in that the common interpretations of *p-values* rely on this definition, rather than definitions that involve statements about “probability of observing data even more extreme than what we actually observe” (*p* stands for probability, owing to this definition).

Notice that the *p-value* is a random variable, and for the definition to make sense, the critical regions $S_{1,\alpha}$ need to be nested for different significance levels α (this is typically, but not always, the case).

Consider a test of the form in eq. (1). The critical value needs to be chosen to satisfy the level constraint in eq. (2). If the null is simple, then we know F_0 , the cumulative distribution function (CDF) of $T(X)$ under the null. We can therefore set the critical value to solve $\alpha = P(T(X) \geq c_\alpha) = 1 - F_0(c_\alpha)$, which yields $c_\alpha = F_0^{-1}(1 - \alpha)$. It then follows from the definition that the p -value is given by $p = 1 - F_0(T(X))$. So the p -value is just a transformation of the test statistic to a $[0, 1]$ scale.

Example 1 (continued). Here the null is not simple. However, since the null rejection probability is increasing in θ , it is maximized at θ_0 , and so taking $T(X) = Z$ to be the z -statistic, we can take F_0 to be the standard normal CDF. Consequently, for a generic significance level α , the test takes the form $\phi(X) = \mathbb{1}\{Z \geq \Phi^{-1}(1 - \alpha)\}$. The p -value is therefore given by

$$p = 1 - \Phi(Z) = 1 - \Phi\left(\frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}}\right).$$

What's the distribution of the p -value under the null? ☒

Remark 3. One can show more generally that if the rejection regions $S_{1,\alpha}$ are nested and if $P_\theta(X \in S_{1,\alpha}) = \alpha$ for all $\alpha \in (0, 1)$ and $\theta \in \Theta_0$, then $P_\theta(p \leq u) = u$ for all $\theta \in \Theta_0$ and all $u \in [0, 1]$. In this case, the p -value has a uniform distribution under the null.

As an exercise, think about whether the equality $P_\theta(p \leq u) = u$ still holds, or whether it holds as an inequality if for some α and some $\theta \in \Theta_0$, $P_\theta(X \in S_{1,\alpha}) < \alpha$.

If the p -value is much smaller than 0.05, then our decision is not particularly sensitive to having chosen 0.05 as the significance level. Moreover, reporting the p -value rather than just the decision to reject has an advantage that, once the p -value is reported, any researcher can make their own decision about whether to reject the null, depending on their own favorite level of the test.

Let us now review some frequent misunderstandings associated with p -values:

1. Smaller p -values don't mean that we are more confident in rejecting the null (what does "more confident" even mean?).
2. p -value is not the probability that the null is true. That probability is either 0 or 1, since the null is either true or not, and as frequentists, we don't have probabilistic beliefs about it.
3. p -value is not the probability of falsely rejecting the null. This probability is measured by the size of the test.
4. One minus p -value is not the probability of the alternative being true. Again, there is no such probability since parameters are not random.
5. The level of the test is not determined by a p -value. I hope that's not how you do empirical work!
6. p -value is not the probability that the observed association is due to chance. In fact, the p -value is computed under the null, i.e., assuming any association in the data is due to chance.

If you'd like to see more misinterpretations, read Greenland et al. (2016).

Example 3. Let $X_1, \dots, X_n$ be a random sample from $\mathcal{N}(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$. We want to test $H_0: \sigma^2 = \sigma_0^2$ against the alternative $H_1: \sigma^2 < \sigma_0^2$. Note that both hypotheses are composite since neither of them restricts μ .

The *ad hoc* method dictates that it may be reasonable to construct a test based on sample variance S_n^2 . We know that $(n-1)S_n^2/\sigma^2 \sim \chi^2(n-1)$. Since small values of $(n-1)S_n^2/\sigma_0^2$ are a sign in favor of the alternative, we should reject for small values of this statistic. Under the null, $(n-1)S_n^2/\sigma_0^2 \sim \chi^2(n-1)$. Then a test with level, say, 5%, rejects the null hypothesis if $(n-1)S_n^2/\sigma_0^2 < F_{\chi^2(n-1)}^{-1}(0.05)$ where $F_{\chi^2(n-1)}^{-1}(0.05)$ denotes the 5%-quantile of the $\chi^2(n-1)$ distribution. What is the power of this test? Fix $\sigma_1^2 < \sigma_0^2$. Then

$$\begin{aligned} P_{\sigma_1^2} \left((n-1)S_n^2/\sigma_0^2 \leq F_{\chi^2(n-1)}^{-1}(0.05) \right) &= P_{\sigma_1^2} \left((n-1)S_n^2/\sigma_1^2 \leq (\sigma_0^2/\sigma_1^2) F_{\chi^2(n-1)}^{-1}(0.05) \right) \\ &= F_{\chi^2(n-1)}((\sigma_0^2/\sigma_1^2) F_{\chi^2(n-1)}^{-1}(0.05)), \end{aligned}$$

where $F_{\chi^2(n-1)}$ denotes the cdf of the $\chi^2(n-1)$ distribution. So the power of the test increases as σ^2 decreases. Notice that it doesn't depend on μ , only σ^2 : in general this may not be the case.

Suppose $n = 101$, $\sigma_0^2 = 1$, and we observe $S_n^2 = 0.9$. What is the p -value of our test? By previous arguments, the smallest significance level α for which we'd reject is given by $(n-1)S_n^2/\sigma_0^2 = F_{\chi^2(n-1)}^{-1}(\alpha)$, so the p -value equals

$$p = F_{\chi^2(n-1)}((n-1)S_n^2/\sigma_0^2) = F_{\chi^2(100)}(100 \cdot 0.9/1) = F_{\chi^2(100)}(90) \approx 0.25. \quad \square$$

For t -tests and z -tests, reporting the p -value makes little sense: it is much more informative to report the estimate and the standard error. However, reporting a p -value can be useful when the test statistic is harder to interpret, such as when we conduct an F test. Since most people have a hard time interpreting values of F statistics, it is more informative to transform the F statistic to a p -value scale.

3. PIVOTAL STATISTICS

A statistic is called *pivotal* if its distribution is independent of unknown parameters. For instance, $(n-1)S_n^2/\sigma_0^2$ from Example 3 is pivotal under the null since its distribution does not depend on the *nuisance parameter* μ . Such pivotal statistics are useful, since they allow us to calculate critical values for tests based on these statistics in closed form.

Example 4. Suppose we observe $X_i \sim \mathcal{N}(\mu, \sigma^2)$, and wish to test the null $H_0: \mu = \mu_0$ against the alternative $H_1: \mu \neq \mu_0$. Then the distribution of the t -statistic $T_n = (\bar{X}_n - \mu_0)/\sqrt{S_n^2/n}$ is $t(n-1)$ under the null: it is pivotal. Since, intuitively, large values of $|T_n|$

are evidence against the null, this suggests a test of the form

$$\phi(X) = \mathbb{1}\{|T_n| \geq c\}.$$

Since the probability density function (PDF) of t -distribution is symmetric around 0, the appropriate critical value c is given by $F_{t(n-1)}^{-1}(1 - \alpha/2)$, the $1 - \alpha/2$ quantile of the t -distribution. Indeed,

$$\begin{aligned} P_{\mu_0}(|T_n| \geq F_{t(n-1)}^{-1}(1 - \alpha/2)) \\ &= P_{\mu_0}(T_n < -F_{t(n-1)}^{-1}(1 - \alpha/2)) + P_{\mu_0}(T_n > F_{t(n-1)}^{-1}(1 - \alpha/2)) \\ &= 2P_{\mu_0}(T_n > F_{t(n-1)}^{-1}(1 - \alpha/2)) = \alpha. \quad \square \end{aligned}$$

Example 5. Let $X_1, \dots, X_{n_X}$ and $Y_1, \dots, Y_{n_Y}$ be independent random samples from distributions $\mathcal{N}(\mu_X, \sigma_X^2)$ and $\mathcal{N}(\mu_Y, \sigma_Y^2)$, with unknown σ_X^2 and σ_Y^2 . We want to test the null $H_0: \mu_X = \mu_Y$ against $H_1: \mu_X > \mu_Y$.

A natural place to start is to note that if the null hypothesis is true, then $\bar{X}_{n_X}$ should be close to $\bar{Y}_{n_Y}$ with high probability. But $\bar{X}_{n_X} - \bar{Y}_{n_Y} \sim \mathcal{N}(0, \sigma_X^2/n_X + \sigma_Y^2/n_Y)$ with σ_X^2 and σ_Y^2 unknown. Another possibility is to consider

$$t = \frac{\bar{X}_{n_X} - \bar{Y}_{n_Y}}{\sqrt{S_X^2/n_X + S_Y^2/n_Y}},$$

but this doesn't work either as the exact distribution of $S_X^2/n_X + S_Y^2/n_Y$ is not pleasant:

$$S_X^2/n_X + S_Y^2/n_Y \sim \frac{\sigma_X^2}{n_X} \frac{\chi^2(n_X - 1)}{n_X - 1} + \frac{\sigma_Y^2}{n_Y} \frac{\chi^2(n_Y - 1)}{n_Y - 1}.$$

Indeed, there are no pivotal statistics in this problem, and finding a good test is difficult: it is known as the *Behrens-Fisher problem*, and people thought about this problem since Behrens (1929) and Fisher (1939). People usually appeal to asymptotics and argue that, by the law of large numbers, as $n_X, n_Y \rightarrow \infty$, $S_X^2/\sigma_X^2 \xrightarrow{p} 1$ and $S_Y^2/\sigma_Y^2 \xrightarrow{p} 1$, so that

$$\begin{aligned} \frac{S_X^2/n_X + S_Y^2/n_Y}{\sigma_X^2/n_X + \sigma_Y^2/n_Y} &= \frac{(\sigma_X^2/n_X)(S_X^2/\sigma_X^2) + (\sigma_Y^2/n_Y)(S_Y^2/\sigma_Y^2)}{\sigma_X^2/n_X + \sigma_Y^2/n_Y} \\ &= S_X^2/\sigma_X^2 + \frac{\sigma_Y^2/n_Y}{\sigma_X^2/n_X + \sigma_Y^2/n_Y} (S_Y^2/\sigma_Y^2 - S_X^2/\sigma_X^2) \\ &\xrightarrow{p} 1. \end{aligned}$$

Thus, under the null, by Slutsky's theorem, $t \Rightarrow \mathcal{N}(0, 1)$. So we can use quantiles of standard normal distribution to form a test with size approximately equal to the required level of the test. This gives us a test with "asymptotically correct size".

If n_X or n_Y is small, then the asymptotics may not be accurate. The Behrens-Fisher problem is still relevant today: it's just an easier version of the problem of how to do

finite-sample valid inference in regression that is robust to heteroskedasticity and/or clustering. There is a growing econometric literature considering various solutions. As an aside, two classic approaches to the Behrens-Fisher problem are to use critical values from a t -distribution with $\min(n_X - 1, n_Y - 1)$ degrees of freedom, which yields valid, but conservative tests (Mickey and Brown 1966), or to use a degrees of freedom adjustment due to Welch (1951). ☒

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